3 Demonstrate the working of the decision tree based ID3 algorithm. Use an appropriate data set for building the decision tree and apply this knowledge to classify a new sample.

Code:

```
import pandas as pd
from sklearn.tree import DecisionTreeClassifier, export_text

data = pd.DataFrame({
    'Weather': ['Sunny', 'Sunny', 'Rainy', 'Rainy', 'Overcast', 'Sunny'],
    'Wind': ['Weak', 'Strong', 'Weak', 'Strong', 'Weak', 'Weak'],
    'Play': ['No', 'No', 'Yes', 'No', 'Yes', 'Yes']
})

X = pd.get_dummies(data[['Weather', 'Wind']])
y = data['Play']

model = DecisionTreeClassifier(criterion='entropy', random_state=0)
model.fit(X, y)

print(export_text(model, feature_names=list(X.columns)))
print("Training Accuracy:", model.score(X, y))

sample = pd.DataFrame([{'Weather': 'Rainy', 'Wind': 'Weak'}])
sample = pd.get_dummies(sample).reindex(columns=X.columns,
fill_value=0)
print("Prediction:", model.predict(sample)[0])
```

Output:

